# The Structured Addressability Limit

A note for independent review of the core mathematics

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**Abstract**

We ask how much exponentially growing history a proper metric host can carry under a linear radial budget, when the code is required to preserve successively more of the source's own structure. The kernel is a packing converse, $\beta \le c\,h_{\mathrm{pack}}$. The distinctive claim is that the resulting capacity is not a single number: imposing a quasi-isometry obligation on the code can strictly reduce what is representable, and we exhibit a proper metric host and an explicit source where it does—and, over the family of sources whose own metric has uniformly bounded balls (bounded valence), at the level of capacities: relational capacity 0 against block capacity $\log 2$. The converse, the block endpoint, the separation, the capacity-level separation, and the host's properness and packing entropy are formally verified in Lean 4 against Mathlib. This note states the definitions and results precisely, marks exactly which statements are machine-checked and which are not, and isolates the questions we believe are open.

## 1 Setup

Throughout, $(M, d)$ is a metric space with basepoint $o$, and $\varepsilon > 0$ is a fixed resolution. $S$ is a set of *histories*.

**Definition 1** (Source)**.** A *source* is a map $\sigma : \mathbb{N} \to \mathcal{P}_{\mathrm{fin}}(S)$ with each census $\sigma(R)$ finite and nonempty. It *has growth* $\beta$ if $\log|\sigma(R)|/R \to \beta$.

The source owns its filtration. This matters: bundling the census into the code makes "can this host carry *this* source?" unstatable, and conflates it with "what rates does this host support?"

**Definition 2** (Code)**.** A *code* is a pair $\kappa = (\mathrm{addr}, r)$ with $\mathrm{addr} : \mathbb{N} \to S \to M$ and $r : \mathbb{N} \to \mathbb{R}$. Write $\kappa_R(\sigma) = \mathrm{addr}(R)\big(\sigma(R)\big) \subseteq M$.

**Definition 3** (Obligations)**.** An *obligation* is a predicate on pairs $(\sigma, \kappa)$.

- **Base** = **Faithful** $\wedge$ **Radial**: $\mathrm{addr}(R)$ is injective on $\sigma(R)$, its image is $\varepsilon$-separated, and $\kappa_R(\sigma) \subseteq \bar{B}(o, r(R))$.
- **Relational** at lawful distortion $q = (D, K)$, $D \ge 1$, $K \ge 0$ (requires a metric on $S$): for all $R$ and all $u, v \in \sigma(R)$,

$$D^{-1} d_S(u, v) - K \;\le\; d\big(\mathrm{addr}(R)u, \mathrm{addr}(R)v\big) \;\le\; D\, d_S(u, v) + K.$$

Base feasibility is not optional. The radial budget $r$ is a free field of the code, unrelated *a priori* to where addr places anything; without Base, a code mapping every history to a single point and declaring $r(R) = cR$ satisfies every rate condition, and a one-point host reports a positive rate. (This degenerate witness is included in the formalization precisely so the omission cannot recur.)

**Definition 4** (Representability and capacity)**.** $\sigma$ is *$\mathcal{A}$-representable at radial rate $c$*, written $\mathrm{Rep}_{\mathcal{A}}(\sigma, c)$, if there is a code $\kappa$ with $\mathrm{Base}(\sigma, \kappa)$, $\mathcal{A}(\sigma, \kappa)$, $r(R) \to \infty$, and $r(R)/R \to c$.

For a declared family $\mathfrak{S}$ of sources,

$$\begin{aligned} \mathsf{Rates}_{\mathfrak{S},\mathcal{A}}(c) &= \big\{\beta : \exists\, \sigma \in \mathfrak{S},\ \sigma \text{ has growth } \beta \text{ and } \mathrm{Rep}_{\mathcal{A}}(\sigma, c)\big\}, \\ \mathsf{C}_{\mathfrak{S},\mathcal{A}}(c) &= \sup \mathsf{Rates}_{\mathfrak{S},\mathcal{A}}(c). \end{aligned}$$

Capacity is a set before it is a number: boundedness is a theorem (Theorem 7), and the supremum is taken only afterwards.

**Definition 5** (ULF)**.** $\sigma$ is *uniformly locally finite* if $\exists K > 0$ with $|\sigma(R+1)| \le K\,|\sigma(R)|$ for all $R$.

This is a finite-generation-rate condition on the source's clock (Lean `Source.ULF`); it is *not* a restriction on the universal converse, which holds for arbitrary sources including bursty ones. A second, independent finiteness condition concerns the source's own metric.

**Definition 6** (Metric-ULF)**.** $\sigma$ is *metric-ULF* (Lean `Source.MetricULF`) if there is $f : \mathbb{N} \to \mathbb{N}$ with $|\{v \in \sigma(R) : d_S(u, v) \le t\}| \le f(t)$ for all $R$, all $u \in \sigma(R)$ and all $t$: uniformly bounded balls in $d_S$, the standard meaning of a uniformly locally finite metric space, and bounded valence in the tree case.

The binary tree is both clock-ULF ($K = 2$) and metric-ULF ($f(t) = 2^t$); a star with $2^i$ leaves is clock-ULF and not metric-ULF. The canonical class for relational statements is the metric-ULF clocked tree, and the distinction is exactly what separates Theorem 14 from Question 16.

## 2 The three stacked facts

**Theorem 7** (Packing converse; Lean)**.** *Let $M$ be proper and $\varepsilon > 0$, and suppose the packing growth converges:*

$$\frac{\log P\big(\bar{B}(o, \rho), \varepsilon\big)}{\rho} \longrightarrow h_{\mathrm{pack}} \qquad (\rho \to \infty).$$

*Then for every family $\mathfrak{S}$ and every obligation $\mathcal{A}$,*

$$\beta \in \mathsf{Rates}_{\mathfrak{S},\mathcal{A}}(c) \implies \beta \le c\, h_{\mathrm{pack}}.$$

The finite-depth form, which is what a measurement computes, is $\log|\sigma(R)|/r(R) \le \log P(\bar{B}(o, r(R)), \varepsilon)/r(R)$; the asymptotic statement discards an additive constant.

**Theorem 8** (Obligation monotonicity; Lean)**.** *If $\mathcal{A} \Rightarrow \mathcal{B}$ pointwise then* $\mathsf{Rates}_{\mathfrak{S},\mathcal{A}}(c) \subseteq \mathsf{Rates}_{\mathfrak{S},\mathcal{B}}(c)$. *Likewise $\mathfrak{S} \subseteq \mathfrak{T}$ gives* $\mathsf{Rates}_{\mathfrak{S},\mathcal{A}} \subseteq \mathsf{Rates}_{\mathfrak{T},\mathcal{A}}$.

**Theorem 9** (Block endpoint, constructive; Lean)**.** *Let $M$ be proper, $\varepsilon > 0$, $c > 0$, and* $\log P(\bar{B}(o, \rho), \varepsilon)/\rho \to h_{\mathrm{pack}}$. *Taking $\sigma(R)$ to be an optimal $\varepsilon$-packing of $\bar{B}(o, cR)$ and* $\mathrm{addr}(R) = \mathrm{id}$ *yields* $c\, h_{\mathrm{pack}} \in \mathsf{Rates}_{\mathrm{all},\,\mathrm{noExtra}}(c)$, *hence* $\mathsf{C}_{\mathrm{all},\,\mathrm{noExtra}}(c) = c\, h_{\mathrm{pack}}$.

*Remark* 10 (Depthwise recoding). Theorem 9 achieves the endpoint with codebooks chosen independently at each radius: every address may be rewritten at every depth. It is *not* a nested, persistent, or successively refinable code, and the Shannon analogy should be read with that qualification. Persistent achievability is open.

Together:

$$\beta \;\le\; \mathsf{C}_{\mathfrak{S},\mathcal{A}}(c) \;\le\; \mathsf{C}_{\mathfrak{S},\mathrm{noExtra}}(c) \;=\; c\, h_{\mathrm{pack}}.$$

The content of the theory is entirely in whether the middle inequality can be strict.

**Theorem 11** (The persistent rung; Lean)**.** *Let $M$ be proper, $\varepsilon > 0$, $c > 0$, with convergent packing rates at resolutions $\varepsilon$ and $2\varepsilon$. Over retentive sources ($\sigma(R) \subseteq \sigma(R{+}1)$, `Source.Retentive`) under the obligation that an assigned address never moves (`Stable` $m$, any $m \geq 0$),*

$$c\,h_{\mathrm{pack}}(2\varepsilon) \;\leq\; \mathsf{C}_{\mathrm{pers}}(\varepsilon) \;\leq\; \mathsf{C}_{\mathrm{noExtra}}(\varepsilon) \;=\; c\,h_{\mathrm{pack}}(\varepsilon),$$

*and if the balls of radius $2\varepsilon$ have uniformly bounded packing number at resolution $\varepsilon$, both inequalities are equalities: the block rate is achieved by one nested family of addresses that never move (`Persistent.lean: achievable_stable_persistent, capacity_stable_between, capacity_stable_eq_packing_of_boundedGeometry`).*

The construction is a nested family of separated sets, each extended to a cover of the next ball by a maximal separated subset of the far region, then thinned to a nested census of size exactly the packing count at $2\varepsilon$; the identity map is the code. The equality clause is the scale-invariance theorem of `CoarseConverse`. Nesting therefore costs at most the gap between two scales of packing entropy; the ladder's price, where there is one, is paid at the relational rung.

## 3 The separation

**Definition 12** (Balloon)**.** Let $B$ have points $\mathsf{hub}(i)$, $i \in \mathbb{N}$, and $\mathsf{leaf}(i,a)$, $a \in \mathbb{Z}/2^i$, with

$$d(\mathsf{hub}\,i, \mathsf{hub}\,j) = |i-j|, \quad d(\mathsf{hub}\,i, \mathsf{leaf}(j,b)) = |i-j|+1,$$

$$d(\mathsf{leaf}(i,a), \mathsf{leaf}(j,b)) = \begin{cases} 0 & i=j,\ a=b \\ 2 & i=j,\ a\neq b \\ |i-j|+2 & i\neq j. \end{cases}$$

$B$ is a metric space (triangle inequality verified formally). It is the ray $\mathbb{N}$ with a star of $2^i$ leaves attached at $\mathsf{hub}(i)$; balls are finite, so $B$ is proper, and $h_{\mathrm{pack}}(B, \mathsf{hub}\,0; \varepsilon) = \log 2$ for $\varepsilon \leq 1$.

Let $T_2$ be the infinite rooted binary tree, $S = \{0,1\}^*$, with $\sigma_2(R) = \{0,1\}^R$ and $d_S$ the tree metric via longest common prefix. Then $\sigma_2$ is ULF with $K = 2$ and has growth $\log 2$.

**Theorem 13** (Separation; Lean)**.** *For every lawful distortion $q = (D, K)$, at $\varepsilon = 1$ and $c = 1$,*

$$\log 2 \in \mathsf{Rates}_{\{\sigma_2\},\,\mathrm{noExtra}}(1) \qquad \textit{and} \qquad \log 2 \notin \mathsf{Rates}_{\{\sigma_2\},\,\mathrm{Relational}(q)}(1).$$

*In particular the two rate sets differ.*

*Proof sketch. Block.* Send $\{0,1\}^R$ injectively onto the $2^R$ leaves of $\mathsf{hub}(R)$. Distinct leaves are at distance $2 \geq \varepsilon$, and all lie in $\bar{B}(\mathsf{hub}\,0, R+1)$, so $r(R) = R+1$ works and $r(R)/R \to 1$.

*Relational.* Suppose $\kappa$ satisfies Relational($q$). Choose $L$ with $D^{-1} \cdot 2(L+1) - K > 2$. For $u \neq v \in \{0,1\}^k$, the padded words $u0^L, v0^L \in \{0,1\}^{k+L}$ satisfy $d_S \geq 2(L+1)$, so their images are at host distance $> 2$. In $B$, two points at distance $> 2$ cannot share a hub's star, so the $2^k$ padded images occupy $2^k$ distinct hubs. But the radial budget confines them to $\bar{B}(\mathsf{hub}\,0, r(R))$, which meets only $\lfloor r(R) \rfloor + 1$ hubs. Since $r(R)/R \to 1$, this is $O(R)$, while $2^k$ grows exponentially in $R$. Contradiction. □

The mechanism is that $B$ has exponentially many points but only linearly many *bounded-diameter shells*, whereas $T_2$ has exponentially many mutually tree-distant vertices at each depth. The same count lifts from one source to the natural family.

**Theorem 14** (Capacity-level separation over metric-ULF sources; Lean)**.** *Let $\mathfrak{S}$ be the family of metric-ULF sources. For every lawful distortion $q$, every $\varepsilon > 0$ and every $c$, no source in $\mathfrak{S}$ of positive growth is* Relational($q$)*-representable in $B$ at rate $c$:* $\mathsf{Rates}_{\mathfrak{S},\mathrm{Relational}(q)}(c) \subseteq \{0\}$. *At $\varepsilon = c = 1$ the block rate set contains $\log 2$ and the relational rate set contains $0$, so as numbers*

$$\mathsf{C}_{\mathfrak{S},\mathrm{Relational}(q)}(1) = 0 \;<\; \log 2 = \mathsf{C}_{\mathfrak{S},\mathrm{noExtra}}(1).$$

*Proof sketch.* Two addresses in one star are at host distance $\leq 2$, so the *lower* relational bound gives $d_S \leq D(2+K) =: t_0$ for their histories, and a star holds at most $f(t_0)$ histories of one census. The radial budget meets $\lfloor r(R) \rfloor + 1$ stars, so $|\sigma(R)| \leq f(t_0)(r(R)+1)$, linear in $R$; a source of growth $\beta > 0$ has $|\sigma(R)| \geq e^{\beta R/2}$ eventually. The binary tree is metric-ULF with $f(t) = 2^t$ (generation-$R$ vertices within tree distance $t$ share a prefix of length $R - t$), and Theorem 13 supplies its block code; the one-history source supplies 0. The scalar form uses that $B$ is proper with $h_{\mathrm{pack}}(B; 1) = \log 2$ (now formalized) through the block identity. Only the lower side of the quasi-isometry sandwich is used. □

The balloon is the instance at scale $\varepsilon' \geq 2$ of a general statement.

**Theorem 15** (Coarse packing converse; Lean)**.** *Let $M$ be proper, $\sigma$ metric-ULF with ball bound $f$, and $\kappa$ a code with Base at resolution $\varepsilon$ and* Relational($D, K$)*. Then for every $\varepsilon' > 0$ and every $R$, $|\sigma(R)| \leq f(\lceil D(2\varepsilon' + K) \rceil) \cdot P(\bar{B}(o, r(R)), \varepsilon')$; hence, if $\sigma$ has growth $\beta$ and is relationally representable at rate $c$ and the packing rate at scale $\varepsilon'$ converges to $h_{\mathrm{pack}}(\varepsilon')$, then $\beta \leq c\, h_{\mathrm{pack}}(\varepsilon')$. Relational capacity over metric-ULF sources is bounded by $c$ times the infimum of packing entropy over all scales: the coarse growth rate, a quasi-isometry invariant.*

*Proof sketch.* A maximal $\varepsilon'$-separated subset of the budget ball is an $\varepsilon'$-cover with exactly the packing number of points (Mathlib, `isCover_maximalSeparatedSet`, `encard_maximalSeparatedSet`). Histories assigned to the same cover point have addresses within $2\varepsilon'$, hence source distance at most $D(2\varepsilon' + K)$ by the lower bound, hence at most $f(\lceil D(2\varepsilon' + K) \rceil)$ per cell. In $B$, at any $\varepsilon' \geq 2$, a ball of radius $\rho$ packs at most $\lfloor \rho \rfloor + 1$ points, so $h_{\mathrm{pack}}(B; \varepsilon') = 0$ and Theorem 14 follows (`balloon_relational_rate_nonpos`). □

Block capacity reads packing entropy at the reader's scale; relational capacity reads it at every scale. The gap can open only in hosts without bounded geometry: if every $\varepsilon'$-ball packs at most $N_0$ points at resolution $\varepsilon$, uniformly in the centre, the two packing rates have the same limit (Lean, `packingRate_scale_invariant`). Scale-invariant packing entropy is therefore necessary for zero relational tax over metric-ULF sources; achievability (a quasi-isometric full-rate tree in the host) is the open half, and neither Gromov hyperbolicity nor the absence of conjugate points is necessary for it (a regular tree times a line carries the binary tree at full rate; conjugate points are invisible to quasi-isometry).

## 4 What is and is not established

| Statement | Status |
|---|---|
| Packing converse, finite-depth and asymptotic | Lean |
| Obligation and family monotonicity | Lean |
| Block endpoint $\mathsf{C}_{\mathrm{noExtra}} = c\,h_{\mathrm{pack}}$, constructive | Lean |
| Separation (Theorem 13), all lawful $q$ | Lean |
| Capacity-level separation over metric-ULF sources (Theorem 14), rate sets and numbers | Lean |
| Coarse packing converse (Theorem 15); $h_{\mathrm{pack}}(B;\varepsilon') = 0$ for $\varepsilon' \geq 2$ | Lean (`CoarseConverse`) |
| Isometry invariance of $\mathsf{Rates}$ | Lean |
| $\mathsf{C}_{\mathrm{rel}} = \mathsf{C}_{\mathrm{block}}$ in $\mathbb{H}^n_\kappa$ | paper only, via [6] |
| $h_{\mathrm{pack}}(B;1) = \log 2$; $B$ proper | Lean (`BalloonPacking`) |
| Persistent rung (Theorem 11): $c\,h_{\mathrm{pack}}(2\varepsilon) \leq \mathsf{C}_{\mathrm{pers}} \leq c\,h_{\mathrm{pack}}(\varepsilon)$, equality under bounded geometry | Lean (`Persistent`) |

All formalized statements depend only on `propext`, `Classical.choice`, `Quot.sound`; there are no `sorry`s and no additional axioms. The audit is executed in CI.

## 5 Two honest limitations

**(i) Representability versus capacity.** Theorem 13 is stated over the singleton family $\{\sigma_2\}$, and Theorem 14 lifts it to the metric-ULF family. It does *not* lift to the clock-ULF family. $B$ is itself a tree metric — the leaves of a common hub are at distance 2, exactly their distance through the hub, so no clique edges are present — and the identity embedding of $B$'s own spanning tree, whose census $\{\mathsf{leaf}(R,\cdot)\}$ is clock-ULF with $K = 2$ but not metric-ULF, is an isometry achieving growth $\log 2$. So $\mathsf{C}_{\mathrm{rel}}(B) = \mathsf{C}_{\mathrm{block}}(B) = \log 2$ over the full clock-ULF family, while $\mathsf{C}_{\mathrm{rel}}(B) = 0 < \log 2$ over the metric-ULF family: the separation is a statement about bounded valence.

**Question 16.** Is there a proper metric host with $h_{\mathrm{pack}} > 0$ such that *no* clock-ULF source of growth $c\,h_{\mathrm{pack}}$ is relationally representable at rate $c$? The balloon is not such a host. Equivalently, over sources of unbounded valence, is a capacity-level separation attainable at all, or is some universality theorem in force?

This is the question we would most like an expert opinion on, and we flag that [1] proves a universality statement in a neighbouring regime (every $n$-point metric space has a subset of size $n^{1-\epsilon}$ embedding in an ultrametric at distortion $O(\log(2/\epsilon)/\epsilon)$). That theorem concerns unbounded-degree ultrametrics on a fixed finite set; the balloon's stars are ultrametric and are exactly what [1] would exploit, which is why unbounded valence is the regime in which universality could still be in force, and bounded valence (Theorem 14) is where it is not.

**(ii) The equality endpoint is imported.** $\mathsf{C}_{\mathrm{rel}}(\mathbb{H}^n_\kappa) = c(n-1)\sqrt{\kappa}$ rests on [6] and is not formalized. [7] and [5] give positivity of relational capacity in broad homogeneous classes, not equality; we state the corresponding homogeneity claims as conjectures, not theorems.

## 6 Inventory by module

The kernel, module by module, as compiled on the pinned Mathlib (Lean 4.34.0-rc1). Every declaration depends only on `propext`, `Classical.choice`, `Quot.sound`; each downstream

module ships with a challenge file (the same statements with every proof replaced by `sorry`) and a job file naming its theorems, definitions and permitted axioms. Paper I cites declarations by name in its [LEAN: …] tags.

| File | Certifies |
|---|---|
| `Packing.lean` | Finite-depth count; `logCard_div_radius_le_packingRate` (the measurable form); `convergent_rate_addressability_limit`; `exists_optimal_blockCode`, `card_le_packingCount`; `distinguishable_le_min`; sub-ball fraction. |
| `Constrained.lean` | The profile: `Source`, `Code`, `Obligation`; `Represents` and `Achievable`; monotonicity in obligation and family; `capacity_block_eq_packing` by depthwise recoding; `structured_addressability_limit`; `Source.ULF`. |
| `Balloon.lean` | The balloon host and `balloon_separates`: the ULF binary tree is block-achievable and not relationally representable at $\varepsilon = c = 1$. |
| `BalloonCapacity.lean` | `Source.MetricULF`; `metricULF_not_relational_pos` (no positive relational rate in $B$ at any $\varepsilon$, $c$, $(D, K)$); `balloon_capacity_separation`, `balloon_capacity_relational_le_zero`, `balloon_separates_metricUlf`; `binaryTreeSource_metricULF` ($f(t) = 2^t$). |
| `BalloonPacking.lean` | `ProperSpace Balloon` (closed balls finite); `balloon_packingRate_tendsto` ($h_{\mathrm{pack}}(B; 1) = \log 2$); `balloon_capacity_block_eq` ($= \log 2$), `balloon_capacity_relational_eq_zero`, `balloon_capacity_strict`. |
| `CoarseConverse.lean` | `census_card_le_coarse_packing` (the counting step at any scale $\varepsilon'$, via Mathlib's maximal separated set as a cover); `coarse_packing_converse` ($\beta \le c\, h_{\mathrm{pack}}(\varepsilon')$ for every $\varepsilon'$); `balloon_packingCount_le`, `balloon_coarse_entropy_zero` ($h_{\mathrm{pack}}(B; \varepsilon') = 0$ for $\varepsilon' \ge 2$); `balloon_relational_rate_nonpos`; `packingCount_anti`, `packingCount_le_mul_packingCount`, `packingRate_scale_invariant`. |
| `Persistent.lean` | The persistent rung: `extendSeparated_isCover` (a separated set extends to a separated cover of the ball, without Zorn), `packingCount_two_mul_le_nestedFinset_card` (a cover at $\varepsilon$ is at least a packing at $2\varepsilon$), `persistentSource_retentive`, `recodingCode_base_persistent`; `achievable_stable_persistent` ($c\, h_{\mathrm{pack}}(2\varepsilon)$ with addresses that never move), `capacity_stable_between`, `capacity_stable_eq_packing_of_boundedGeometry` (equality with the block capacity under bounded geometry), `persistent_chain`. |
| `Rigidity.lean` | `maximal_separated_isNet`; `quasiIsometric_of_net`. |
| `AuditFactors.lean` | `Chain.efficiency_eq_product`, `factors_le_one`, `efficiency_le_one`; `card_chain`, `log_card_chain` (the finite form). |
| `Capacity.lean`, `StateEquation.lean` | Real algebra downstream of $\beta \le c\, h_{\mathrm{pack}}$: efficiency, curvature floor, gauge invariance; the equality-case face `normalized_state_equation`; mismatch diagnostics explicitly *not* a Lyapunov theorem. |
| `Window.lean` | For the companion: the $q$-state symmetric channel and its powers; the identity price $Np^L \le \delta$ for $L \ge \log(N/\delta)/(-\log p)$; the Kesten–Stigum line $b\lambda^2 > 1 \iff \beta > 2\rho$ in rate form; exactness of the line as the named hypothesis `ReconstructionThreshold`; the distance reader's window: `identified_of_sites`, `identified_mono`, `readable_branch_lower`, `readable_depth_le`, `window`. |
| `Heredity.lean` | The room of compositions, the reader-side counterpart of the coarse packing converse: carriers of length $L$ over an alphabet of size $a$ have $a^L$ ordered states but at most $(L+2)^{a^k}$ distinct $k$-mer spectra (`card_spectra_le`); histories a code tells apart by composition alone are bounded by that polynomial room (`histories_le_composition_room`); for every fixed $k$ the spectrum map is non-injective on all sufficiently long carriers (`eventually_spectrum_not_injective`). The companion's composition results (its §3.3) rest on this count. |

## 7 Relation to neighbouring work

Kolmogorov–Tikhomirov $\varepsilon$-capacity is the $\mathcal{A} = \text{noExtra}$ case without a clock; the radial budget and the generative clock are what make this a rate problem rather than a counting problem. Buyalo–Schroeder [2] treat trees as coordinates and hosts as domains; we reverse the variance. Successive refinement [4] is the information-theoretic analogue of the persistence obligation, which we do not compute here. [3] gives a rate–distortion AEP for tree-indexed processes, with no host geometry.

## References


[1] Y. Bartal, N. Linial, M. Mendel, A. Naor. *On metric Ramsey-type phenomena*. Ann. of Math. **162** (2005), 643–709.

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